



厦门大学《有机化学 B》课程试卷

化工、药学、环科、生物, 2013 秋季

主考教师:

学生姓名:

(请勿在试卷上作答, 答案请写在专用答题纸上)

一、选择题 (12 小题, 共 22 分)

1. 写出下列化合物在与乙胺发生胺解时的反应活性顺序? (从大到小, 2 分)

(a) CH_3CONH_2 (b) CH_3COCl (c) CH_3COOEt (d) $(\text{CH}_3\text{CO})_2\text{O}$

2. 比较下列羧酸的酸性强弱? (从大到小, 2 分)

(a) $\text{CH}_3\text{CH}_2\text{COOH}$ (b) FCH_2COOH (c) $\text{CH}_3\text{OCH}_2\text{COOH}$ (d) ClCH_2COOH

3. 比较下列化合物活泼氢的酸性强弱? (从大到小, 2 分)

(a) $\text{CF}_3\text{CH}_2\text{OH}$ (b) $\text{CBr}_3\text{CH}_2\text{OH}$ (c) $\text{CCl}_3\text{CH}_2\text{OH}$ (d) $\text{CH}_3\text{CH}_2\text{OH}$

4. 写出下列化合物发生亲核加成反应时活性顺序? (从大到小, 2 分)

(a) $\text{CH}_3\text{CH}_2\text{CHO}$ (b) CF_3CHO (c) CH_3CHO (d) CH_3COCH_3

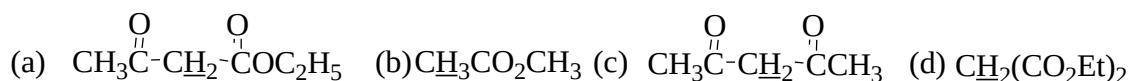
5. 写出下列化合物的碱性强弱顺序? (从大到小, 2 分)

(a) 氨 (b) 三乙胺 (c) 对硝基苯胺 (d) 苯胺

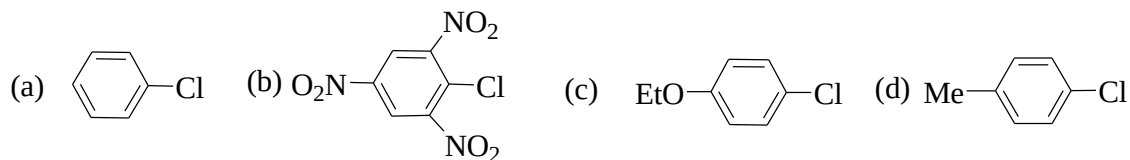
6. 写出下列化合物的沸点高低顺序? (从大到小, 2 分)

(a) 正丁醇 (b) 1-丁酸 (c) 1-氯-丁烷 (d) 正丁烷

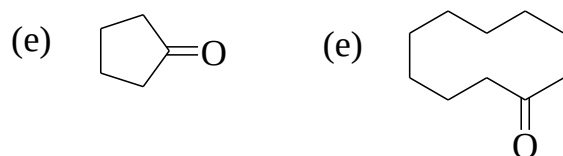
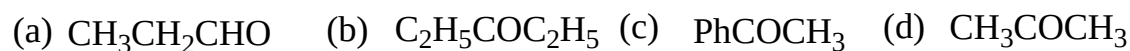
7. 下列哪个化合物有下划线的 α -H 的酸性最强? (2 分)



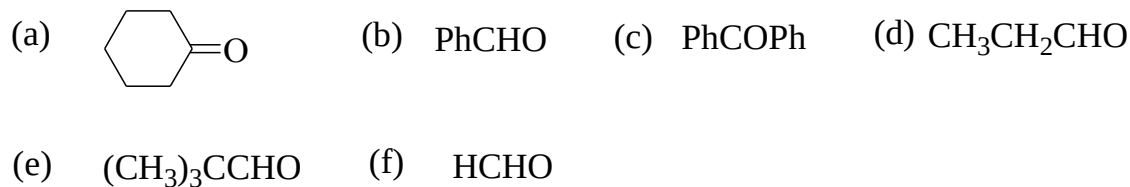
8. 下列哪个化合物可以直接碱水解制备相应的酚类化合物? (2 分)



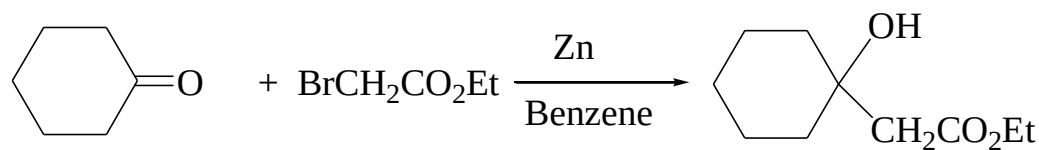
9. 下列化合物中, 哪几种可以和 NaHSO_3 发生反应? (多选, 2 分)



10. 下列化合物中, 哪几种可以在浓碱发生 Cannizzaro 歧化反应? (多选, 2 分)

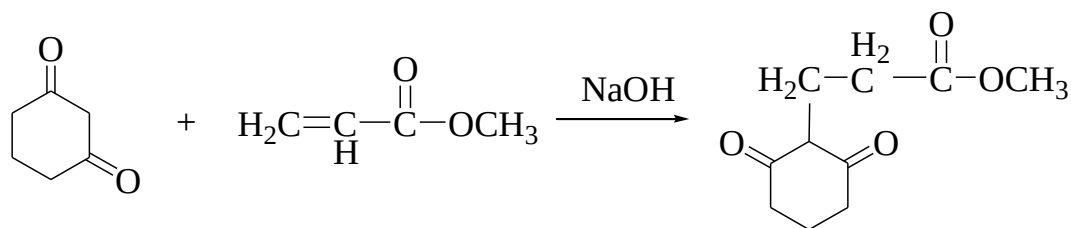


11. 以下反应属于哪一个人名反应? (1 分)



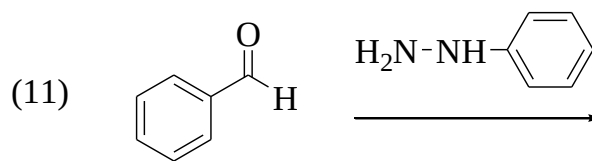
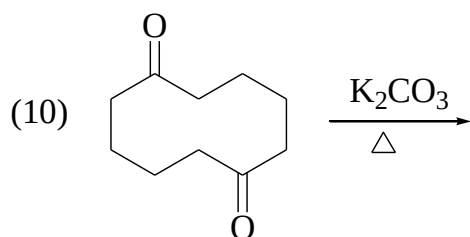
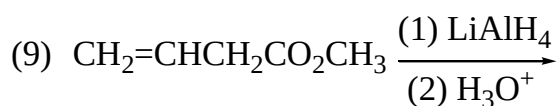
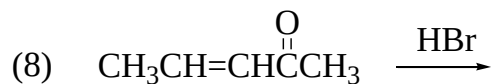
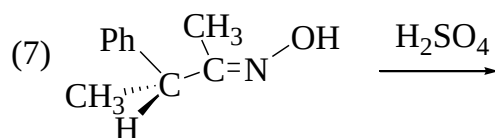
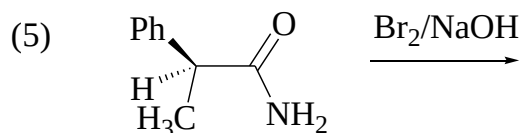
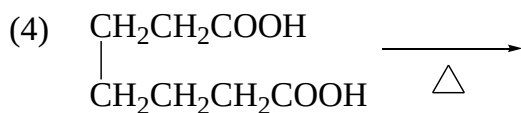
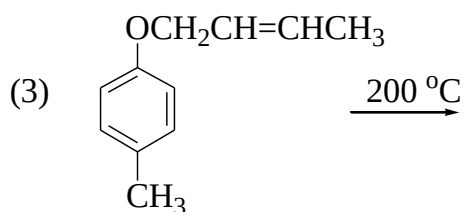
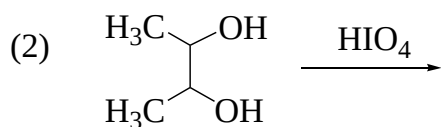
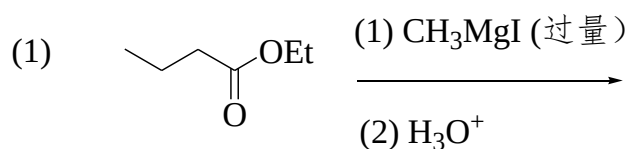
(a) Reformatesky 反应 (b) Claisen 缩合 (c) Micheal 加成 (d) Perkin 反应

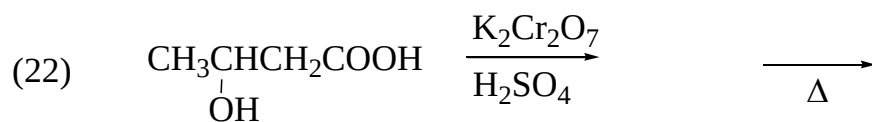
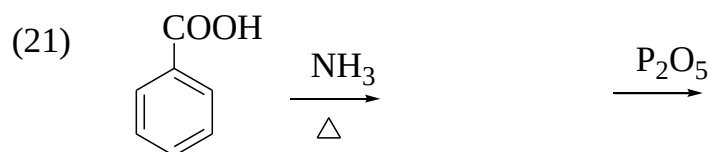
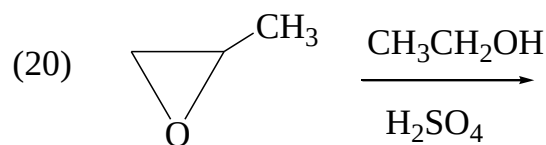
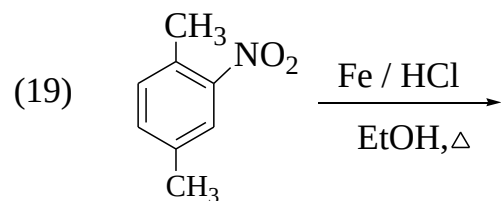
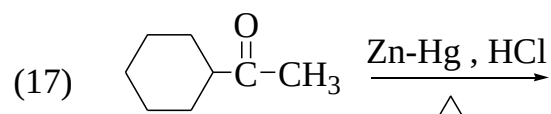
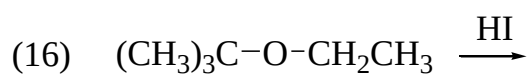
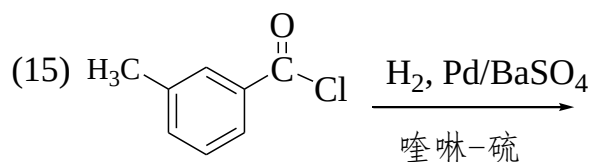
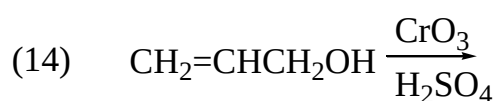
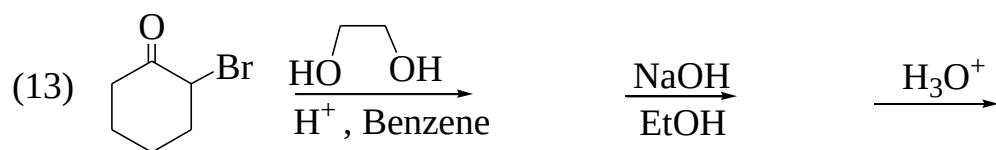
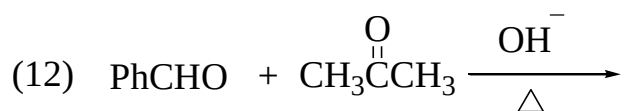
12. 以下反应属于哪一个人名反应? (1 分)



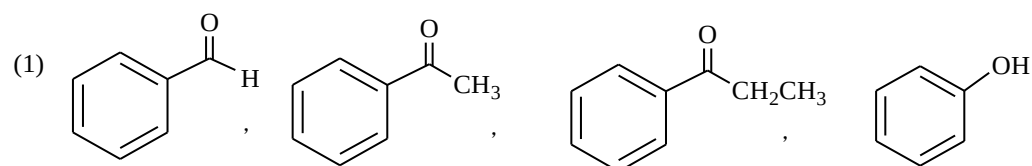
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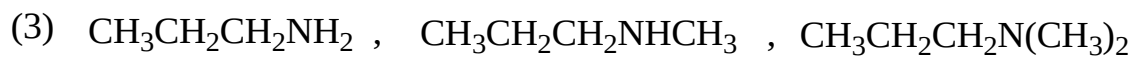
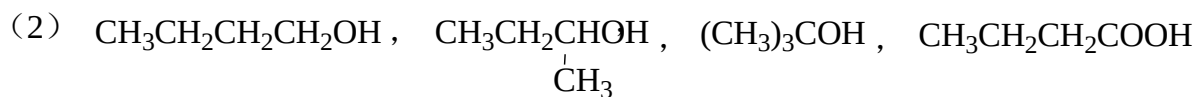
二、请写出下列反应的主要产物（22 小题，共 30 分）



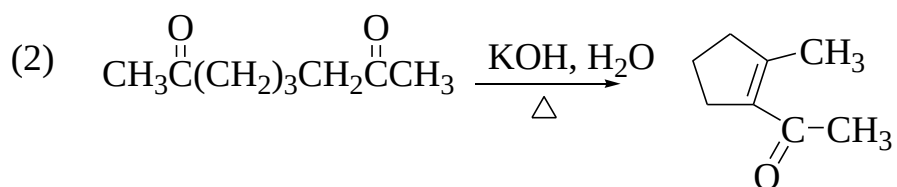
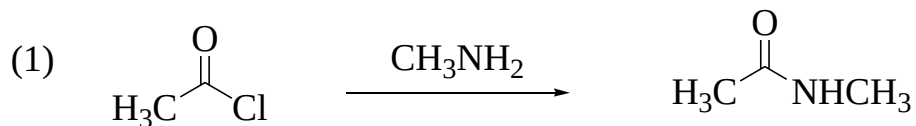


三、请用化学方法鉴别下列各组化合物（3 小题，共 18 分）

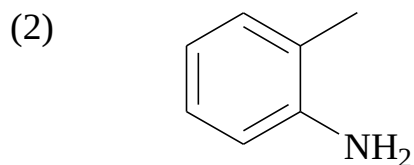
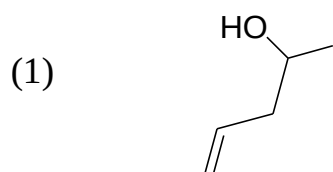




四、请完成下列反应并写出反应机理（2 小题，共 12 分）



五、请用 **C2** 以下(包括 **C2**)的有机物、苯、甲苯及无机试剂合成下列化合物（3 小题，共 18 分）



(3) 以“三乙”（原料要合成）合成法合成以下产物

